

Regarding entropy which statement is False.
(Hint: S is a collection of positive and negative examples)

Select one:

- ☐ a. Entropy measures the uncertainty.
- ☐ b. Entropy is 1 when S contains an equal number of positive and negative examples
- ☒ c. Entropy is 0 when S contains an equal number of positive and negative examples
- ☐ d. Entropy is measured in bits.

6. What is the name of this task that can be solved with machine learning: "The computer program is asked to specify which of k categories some input belongs to"

Select one:

- ☐ a. Generalization
- ☐ b. crossvalidation
- ☐ c. Regression
- ☒ d. Classification

40. What is a membership function of a fuzzy set?

Select one:

- ☐ a. It is a set of elements that have some common property (such as 'Hot')
- ☐ b. A collection of fuzzy sets
- ☒ c. It describes the degree of membership that an object belongs to the set.
- ☐ d. none

24. What is perceptron in Neural network

Select one:

- ☒ a. It is a single layer feed-forward neural network with pre-processing
- ☐ b. It is a neural network that contains feedback
- ☐ c. It is an auto-associative neural network
- ☐ d. It is a double layer auto-associative neural network

Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

Find $A \cap B = ?$

Select one:

- ☐ a. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
- ☒ b. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☐ c. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☐ d. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$

17. In Naive Bayes classification:

Select one:

- ☐ a. Assumption that all input attributes are related to each other
- ☐ b. Assumption that all input attributes are conditionally independent
- ☐ c. Assumption that all input attributes are conditionally dependent
- ☐ d. Assumption that all input attributes are mutually dependent

Question 14

Not yet answered

Marked out of 1.0

Flag question

27. In this learning method, those neurons which respond strongly to input stimuli have their weights updated

Select one:

- ☐ a. Stochastic Learning
- ☐ b. Hebbian Learning
- ☐ c. Gradient Descent Learning
- ☒ d. Competitive learning (SOM)

Question 15

Not yet answered

Marked out of 1.0

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1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A \cap B = ?$

Select one:

- ☒ a. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$
- ☐ b. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☐ c. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
- ☐ d. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$

Question 16

Not yet answered

Marked out of 1.0

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35. If x is A then y is B else y is C . The output of the given fuzzy rule is

Select one:

- ☒ a. a crisp set
- ☐ b. a membership function
- ☐ c. a fuzzy set
- ☐ d. a fuzzy relation

Question 17

Not yet answered

Marked out of 1.0

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37. Defuzzification is done to obtain

Select one:

- ☒ a. Precise fuzzy value
- ☐ b. None of the above
- ☐ c. Crisp output
- ☐ d. The best rule to follow

Question **18**

Not yet answered

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44. What Is The Purpose of Aggregation (in Fuzzy Inference System)?

Select one:

- ☒ a. To gather all the different fuzzy set outputs and combine them into a single fuzzy set outputs
- ☐ b. To gather all the different fuzzy set outputs and average them out to get a single value
- ☐ c. To subtract all the output fuzzy set values from the input values
- ☐ d. To gather all the possible inputs and use the average to gain an output

Question 19

Not yet answered

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30. A fuzzy set A is closed if:

Select one:

- ☐ a. Limit of infinity side of membership function is 1, the other limit is 0.
- ☐ b. Limit of infinity side of membership function is 0, the other limit is 1.
- ☒ c. Limit of both sides of membership function are 0
- ☐ d. Limit of both sides of membership function are 1

Question **20**

Not yet answered

Marked out of 1.0

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29. In unsupervised learning, training set of data includes

Select one:

- ☒ a. Input
- ☐ b. Both input and output
- ☐ c. None
- ☐ d. Output

Question 21

Not yet answered

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5. What is the name of this task that can be solved with machine learning: "In this type of task, the computer program is asked to predict a numerical value given some input."

Select one:

- ☐ a. Regression
- ☐ b. Classification
- ☒ c. Generalization
- ☐ d. Crossvalidation

What is the Bandwidth of fuzzy set A ? The set is given as follow:

$A = (10, 0.1), (15, 0.2), (20, 0.5), (25, 0.4), (30, 0.4), (35, 0.5), (40, 0.2), (45, 0.1)$

Select one:

- ☐ a. 35
- ☐ b. 15
- ☐ c. 15
- ☐ d. 20